Mini-project 1 in Advanced Machine Learning (02460)

by Marcus Christoffersen (s224750), Bob (s271828) & Eve (s314159) on 05 March 2026

Part A: VAE Priors

We train VAEs on binarised MNIST with three priors: standard Gaussian, a 10-component MoG, and a flow-based prior (8 affine coupling layers). For non-Gaussian priors the KL is estimated via Monte Carlo. Figure 1 shows t-SNE projections of prior and aggregate posterior samples. The Gaussian prior forms a unimodal blob misaligned with the multi-modal posterior; MoG and Flow priors learn to better align with the digit-class structure. Test ELBOs (mean±std, 5 runs): Gaussian -88.77 ± 0.69, MoG -88.16 ± 0.67, Flow -88.05 ± 0.38. Differences are small, likely because the decoder capacity is the binding constraint: once the reconstruction term is saturated, prior flexibility yields diminishing returns. The flow prior also shows lower variance, suggesting a more stable training objective.

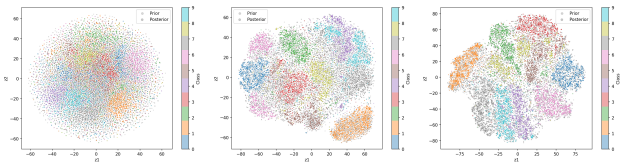

Figure 1: Prior vs. aggregate posterior for Gaussian (left), MoG (mid), Flow (right).

Part B: Sampling Quality

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Figure 2: 2×2 samples: VAE MoG (left), DDPM U-Net (mid), Latent DDPM $\beta = 10^{-3}$ (right).

Table 1: FID (↓) and sampling speed on standard MNIST ($N = 1000$).

Model	FID	Samp./s
VAE (Gaussian)	42.30	13,179
DDPM (U-Net)	36.96	35
Lat. DDPM $\beta = 10^{-6}$	115.15	8,405
Lat. DDPM $\beta = 10^{-3}$	81.91	8,613
Lat. DDPM $\beta = 0.1$	88.80	8,584
Lat. DDPM $\beta = 1$	153.18	8,861

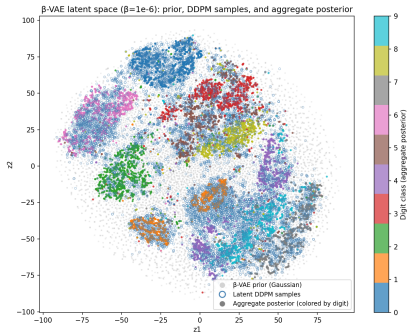

Figure 3: Latent space ($\beta = 10^{-6}$): Gaussian prior (grey), DDPM latents (blue), aggregate posterior (coloured).

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Code snippets

Declaration of use of generative AI

This declaration **must** be filled out and included as the 3rd page of the document. The questions apply to all parts of the work, including research, project writing, and coding.

- I/we have used generative AI tools: [yes / no]

If you answered *yes*, please complete the following sections. List the generative AI tools you have used:

-

Describe how the tools were used:

What did you use the tool(s) for?

At what stage(s) of the process did you use the tool(s)?

How did you use or incorporate the generated output?

References